MSDS 422 | Final Project

Predicting NFL Game Outcomes Using Machine Learning

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Northwestern University · 2025

Project Objective

Research Question

Can we use historical team performance metrics to accurately predict the outcome of an NFL game (win/loss)?

Sports Analytics

Apply modern ML techniques to real-world sports prediction problems

Win/Loss Prediction

Build binary classifiers using team-level historical game statistics

Model Comparison

Evaluate multiple algorithms to identify the best approach for this domain

Data Source — nflverse Repository

DATA ORIGIN

- Open-source nflverse GitHub repository
- Structured, game-level NFL statistics
- Multiple NFL seasons of historical data
- Offensive & defensive team metrics per game
- Play-level granularity available
- Cleaned and engineered for ML modeling

Dataset Snapshot

~2,500+	8	2024
Games	EWMA Features	Hold-out Season

Target Variable

home_win — Binary variable (1 = home team wins, 0 = home team loses)

Baseline accuracy (always predict home win): 56.4%

Feature Engineering — EWMA Metrics

Exponentially Weighted Moving Average (EWMA)

Rather than simple season averages, EWMA gives more weight to recent games — capturing current team form and momentum rather than outdated performance.

🏠 HOME TEAM FEATURES

home_ewma_rushing_offense
Home rushing yards gained per game

home_ewma_passing_offense
Home passing yards gained per game

home_ewma_rushing_defense
Home rushing yards allowed per game

home_ewma_passing_defense
Home passing yards allowed per game

✈️ AWAY TEAM FEATURES

away_ewma_rushing_offense
Away rushing yards gained per game

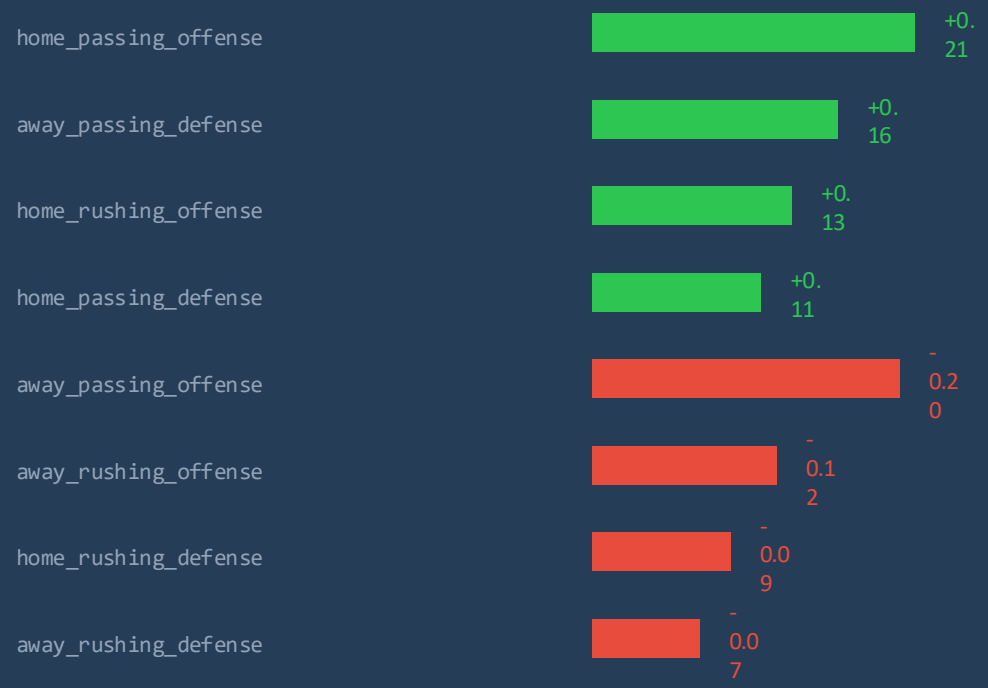
away_ewma_passing_offense
Away passing yards gained per game

away_ewma_rushing_defense
Away rushing yards allowed per game

away_ewma_passing_defense
Away passing yards allowed per game

Exploratory Data Analysis

Feature Correlation with home_win



Key EDA Findings

- Distributions are approximately normal with limited outliers
- Some features are left-skewed due to extreme performance games
- Passing metrics show stronger correlations with win outcomes than rushing
- No single feature dominates — multiple predictors are needed
- Home team features positively correlated; away features negatively correlated
- Home win rate ≈ 56.4% — serves as baseline accuracy

Modeling Approach

Train on seasons prior to 2024 → Hold out 2024 season as final test → Use Leave-One-Season-Out (LOGO) cross-validation

Logistic Regression	Random Forest	Decision Tree	Gradient Boosting
Why chosen:	Why chosen:	Why chosen:	Why chosen:
Linear baseline; interpretable coefficients; good benchmark	Ensemble of trees; captures non-linear patterns; reduces overfitting	Interpretable splits; single tree; useful for understanding key features	Sequential error correction; handles complex feature interactions
Complexity: Low	Complexity: Medium	Complexity: Low-Med	Complexity: High

Evaluation metrics: Accuracy · F1 Score (weighted) · Log Loss · Leave-One-Group-Out Cross-Validation

Model 1 — Logistic Regression

Overview

- Linear classifier for binary outcomes
- Predicts probability of home win via logistic function
- Features scaled with StandardScaler before training
- Serves as interpretable baseline model
- GridSearchCV used to tune regularization (C, penalty, solver)
- Best params: C=0.013, L1 penalty (heavy regularization)
- Heavy regularization suggests features aren't strongly predictive individually

0.67	0.71	0.56
Test Accuracy	F1 Score	Baseline

Cross-Validation Insight

LOGO CV accuracy: ~0.665 (leave-one-season-out)
High log loss indicates model makes confident wrong predictions.
Confusion matrix reveals poor home-loss prediction.
Being linear, it can only draw straight decision boundaries — *likely the main constraint.*

Model 2 — Random Forest ★ Best Overall Performer

Overview

- Ensemble of 200+ decision trees
- Each tree trained on a bootstrap sample
- Random feature subsets at each split (sqrt, log2 tested)
- Captures non-linear relationships in game data
- class_weight='balanced' to handle class imbalance
- Hypertuning: max_depth, n_estimators, min_samples_leaf
- Best: max_depth=20, n_estimators=500, min_samples_leaf=20

0.68	0.72	1.00
Test Accuracy	F1 Score	Train Acc.

Why It Performed Best

Aggregating many trees reduces variance significantly.
Non-linear boundaries capture interaction effects between home & away stats.
Balanced class weights improve recall for minority class (home losses).
Most stable cross-validation performance across seasons.

Model 3 — Decision Tree

Overview

- Single hierarchical tree splitting on feature thresholds
- Initially no depth limit → severe overfitting
- GridSearchCV tuned: max_depth, min_samples_leaf, criterion
- Best params: max_depth=5, min_samples_leaf=50, entropy
- Provides visual interpretability of decision paths
- Useful for understanding which features drive splits
- class_weight='balanced' applied to handle imbalance

⚠ Overfitting Detected

Training Accuracy: 1.00
Test Accuracy: ~0.54

Perfect training fit = pure memorization
Poor generalization to unseen 2024 games

After Hypertuning (max_depth=5)

0.54	0.58
Test Accuracy	F1 Score

Still underperforms ensemble methods — single trees cannot match aggregated wisdom

Model 4 — Gradient Boosting

Overview

- Builds trees sequentially — each tree corrects prior errors
- Learning rate controls the contribution of each tree
- Better generalization than single decision tree
- Sensitive to hyperparameter tuning
- Tuned: n_estimators, learning_rate, max_depth, min_samples_leaf
- Best: n_estimators=200, lr=0.05, max_depth=3, min_samples_leaf=20
- Shallow trees (depth=3) prevent overfitting

0.66	0.71	0.68
Test Accuracy	F1 Score	Best: RF Acc.

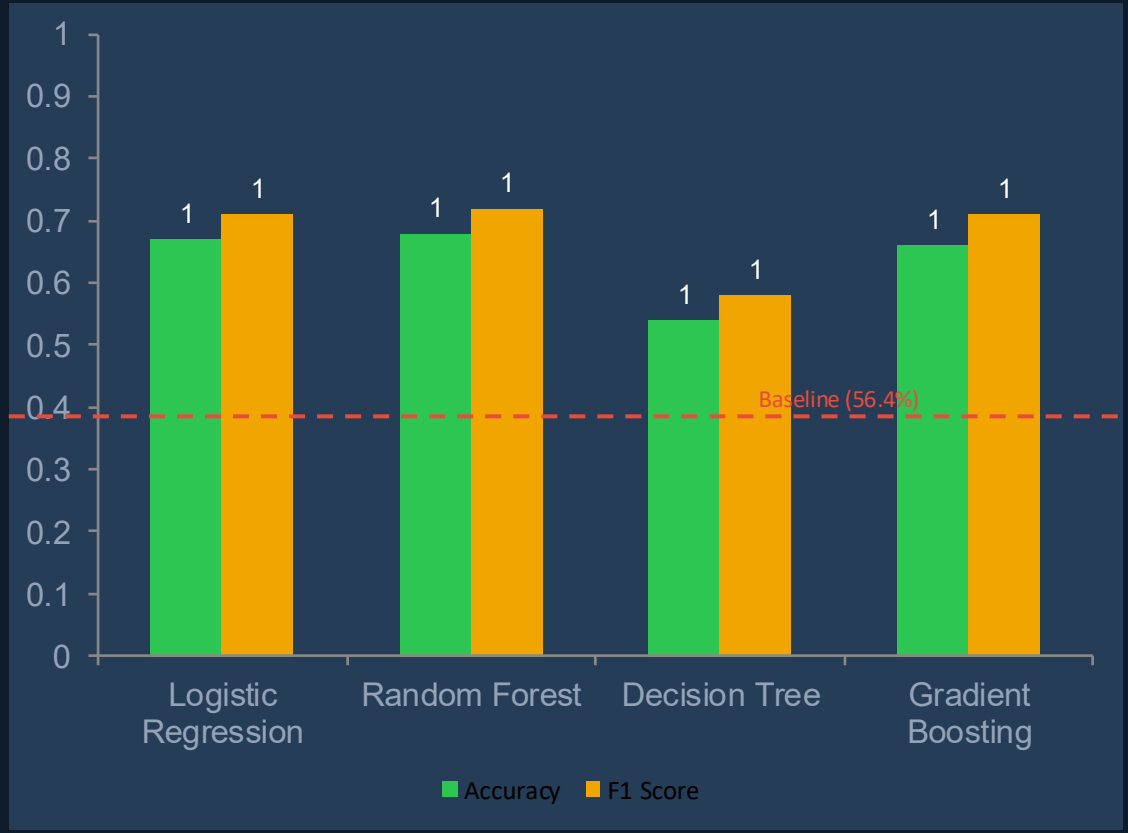
Boosting vs. Bagging Tradeoff

Boosting (GB): Sequentially reduces bias but risks overfitting with high lr or depth.

Bagging (RF): Reduces variance by averaging independent trees.

For NFL game prediction, variance reduction (RF) outperformed bias reduction (GB).

Model Comparison



Results Summary

Model	Acc.	F1
Log. Reg.	0.67	0.71
Rand. Forest ★	**0.68**	**0.72**
Dec. Tree	0.54	0.58
Grad. Boost	0.66	0.71

★ Random Forest achieved the best accuracy and F1 across all models.

Key Findings

Passing > Rushing

Passing offense EWMA showed the highest correlation with home win outcomes, consistent with the modern pass-heavy NFL era.

Ensemble Methods Win

Random Forest and Gradient Boosting both outperformed the single Decision Tree, demonstrating the value of model aggregation.

No Dominant Feature

No single metric explains outcomes. Multiple offensive and defensive metrics collectively contribute to prediction accuracy.

EWMA Adds Value

Exponentially-weighted metrics better capture current team form than season-long averages, improving model generalization.

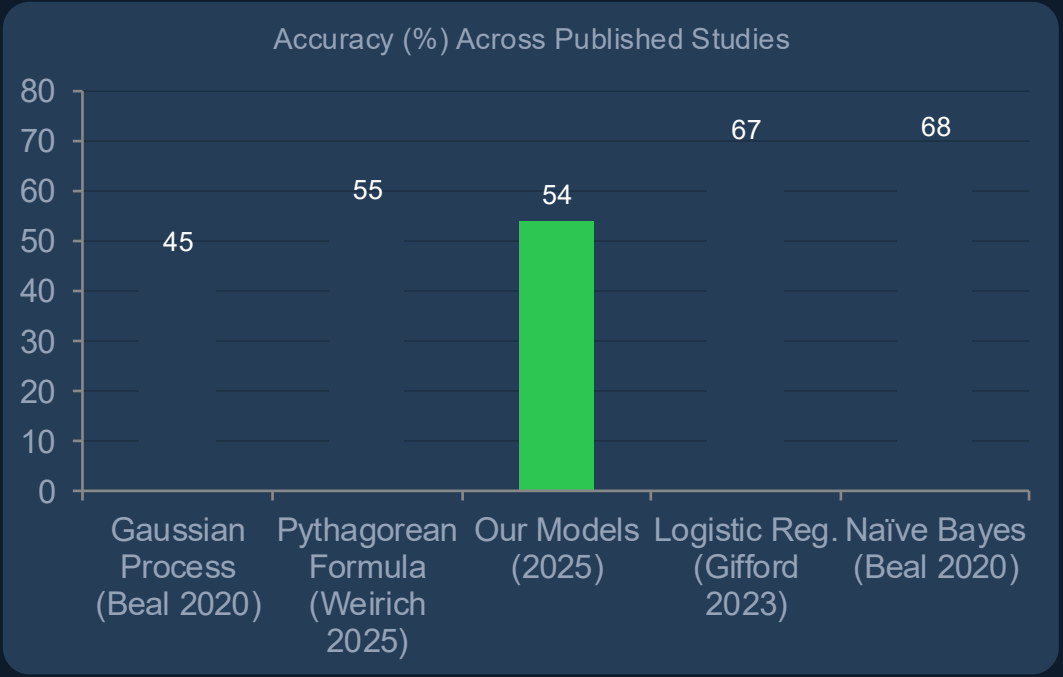
Beat the Baseline

All models except the overfit Decision Tree exceeded the 56.4% baseline of always predicting a home team win.

Inherent Randomness

NFL outcomes contain inherent unpredictability — even the best model tops out near 68%, consistent with prior research.

Literature Context



Our 68% (RF) falls within the top range of published results

Beal et al. (2020)

Nine ML classifiers on NFL data (5 seasons)

Accuracy ranged 44–67% across classifiers

Gifford & Bayrak (2023)

Decision Tree & Logistic Regression for NFL outcomes

Logistic regression competitive with tree methods

Weirich et al. (2025)

Pythagorean formula vs. RF and neural networks

Points scored/allowed most important predictors

Brandon (2024)

ML for NFL point spreads prediction

Ensemble models outperform linear regression

Limitations

Player Injuries

Injury status of key players (QB, star receivers) significantly affects game outcomes but was not included in the feature set.

Weather & Conditions

Outdoor game weather (rain, cold, wind) can suppress passing offenses — an important factor not captured by historical EWMA stats.

Roster Changes

Free agency signings, trades, and draft picks change team composition significantly between and within seasons.

Inherent Randomness

NFL games have high variance: fumbles, deflections, and referee calls introduce outcomes that no model can predict from statistics alone.

Limited Feature Set

Only 8 EWMA features used. Turnover rates, red zone efficiency, third-down conversion, and special teams stats could improve predictions.

Temporal Drift

Team quality can shift dramatically mid-season. A single EWMA decay parameter may not optimally capture slow vs. fast-changing teams.

Future Work & Next Steps

Phase 1
Expand Feature Set

- Add injury report data
- Include betting lines (market consensus)
- Weather conditions for outdoor stadiums
- Red zone & third-down efficiency stats

Phase 2
Advanced Models

- Deep neural networks (LSTM for sequences)
- Player-level tracking data
- Transformer-based models on play sequences
- Bayesian approaches for uncertainty quantification

Phase 3
Production System

- Real-time prediction pipeline
- Live game win-probability updates
- Week-by-week EWMA recalculation
- Automated model retraining each season

Expected improvement: 68% → 72–75% accuracy with player-level data and ensemble stacking

Conclusion

Machine learning models can provide moderate predictive power (~68%) for NFL game outcomes using historical EWMA team performance metrics — consistently outperforming the 56.4% home-win baseline.

0.68	0.72	4	+12%
Best Accuracy (Random Forest)	Best F1 Score (Random Forest)	Models Compared	Gain over Baseline

- Random Forest achieved the best overall performance across accuracy and F1
- Gradient Boosting and Logistic Regression also performed competitively
- Results align with the 44–67% range found in published NFL prediction research
- Richer feature sets (injuries, weather, player data) are the clearest path to improvement

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